

Melanie Valdez

timothy90@example.net 652.611.8968

Professional Summary

AI Researcher with 14 years in Data/AI/ML, currently at Lumen Works. I work at the point where architecture meets delivery: setting technical direction, then staying close enough to the code to know whether it held. Strongest in NumPy, Snowflake and Spark.

Academic Background

Bachelor, Data Science

West Gabriella Institute of Technology | 2011

Employment History

AI Researcher

Lumen Works | May 2015 – Mar 2026 | North Scott

- Migrated ad-hoc notebooks into versioned Python pipelines with reproducible runs and tracked experiments.
- Reduced label noise by redesigning the annotation guidelines, improving validation accuracy by 15%.
- Built an end-to-end training pipeline on Spark, cutting retraining from days to 15 hours.
- Built the feature store that 6 downstream models now read from, ending duplicated feature logic.
- Cut inference cost by 45% through quantisation, batching and autoscaling on the in-house toolchain.

Data Scientist

BrightPath Solutions | Apr 2014 – May 2015 | Simonland (Hybrid)

- Translated ambiguous business questions into a modelling roadmap with product and finance stakeholders.
- Mentored 6 analysts moving into modelling work, with weekly paper reading and code review.
- Trained and deployed NumPy models serving 50M predictions per day at under 60ms p95.
- Wrote the team's model documentation and review standard, now required before any production launch.

Fenwick Digital — Junior Data Scientist

May 2011 – Nov 2013 | West Allison

- Productionised demand forecasting models that reduced stock-outs by 40% across 6 regions.
- Developed an NLP classification system in Python reaching 88% F1, replacing a manual review queue.
- Established offline/online evaluation parity, catching 40 silent regressions before release.

Technical Proficiencies

NumPy (4 years), Snowflake (3 years), Spark (9 years), MLflow (11 years), Airflow (11 years), Databricks (1 year), Kafka (14 years), Python (4 years), SQL (10 years), MongoDB (13 years), Looker (7 years), Vue.js (11 years), Git (6 years), Elasticsearch (9 years)

Selected Projects

Technical blog

Writing up Python experiments and failures, roughly 9k readers a month.

Charity chatbot

Built a NumPy assistant for a local charity to answer volunteer questions; retired it once they moved to a supported platform.

Certificates

AWS Certified Developer - Associate — Amazon Web Services, 2019

HashiCorp Certified: Terraform Associate — HashiCorp, 2019

Selected Publications

"Drift detection for production forecasting models" - accepted at a peer-reviewed workshop, 2013.

Personal Interests

Sailing, Travel photography, Cycling